Web and Social Media Search and Analysis

Course Project Report

Coordinated Propaganda Networks

in Russian-Language Telegram

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1. Objective

This project investigates the ecosystem of Russian-language Telegram channels. Telegram is the dominant platform for political communication in the Russian-speaking world, hosting state media, pro-government "military correspondents" (voenkory), independent and exiled opposition outlets, and a large grey zone of anonymous commentary channels. Because the platform combines mass broadcasting with frictionless re-publishing (forwarding), it is an ideal substrate for coordinated information operations.

Our aim is: (i) to reconstruct the inter-channel network from observable behavior (forwarding, shared links, synchronous posting); (ii) to detect communities and signs of coordinated inauthentic behavior (CIB); and (iii) to characterize how the discourse of these communities differs in tone, emotion, named entities, and topics. The analysis is organized around four research questions:

RQ1. What communities exist within the network of Russian political channels, and do they correspond to political alignment?

RQ2. Are there measurable signs of coordinated behavior - synchronous posting, shared URLs, near-duplicate text, or mass forwarding?

RQ3. How does content (sentiment, emotion, topics, named entities) differ between clusters?

RQ4. Which channels act as bridges between otherwise separate information bubbles?

2. Data

2.1 Seed list construction

We compiled a seed list of political channels using TGStat rankings and keyword search (politics, news, Russia, Ukraine, war, mobilisation), targeting a balanced spread across the political spectrum. Each channel was manually labelled with a political lean - pro (state-aligned), anti (opposition), mixed, or neutral (wire/news services) - and a finer sub-category (e.g. voenkor, state official, independent media, opposition, anonymous political). The labelled list contains 180 candidate rows, of which 167 had a verified public username and were retained for scraping.

During labelling we identified three cases of deliberate mimicry that were resolved by reading the actual content rather than the channel's self-presentation: a channel branded as a "fact-checker" whose output is in fact state-aligned counter-propaganda, and two channels mimicking Ukrainian commentary that are in practice Russian information operations. Two further channels (Girkin/Strelkov and Sobchak) were labelled mixed as methodologically valuable cases of, respectively, anti-Putin far-right and "controlled-opposition" positioning. This motivated a two-pass labelling scheme based on content verification.

2.2 Collection method

Rather than the Telegram API (which requires an account and is subject to rate limits and account bans), we collected data by parsing the public web preview pages t.me/s/<channel>. A custom multi-threaded scraper (wsa_scraper.py) walks each channel from newest to oldest using the ?before= cursor, parsing message id, timestamp, text, view count, forwarding source, reply target, media flags, and all embedded URLs out of the page HTML. The scraper is resume-safe (per-channel cursor stored in SQLite), uses a rotating proxy pool and user agents, and exponential back-off on HTTP 429. Storage uses SQLite in WAL mode with three tables - channels, messages, urls.

The collection window spans six months (23 November 2025 – 23 May 2026). Of the 167 channels, 148 were scraped to completion and 19 partially (their public preview history is shorter than the window). The raw corpus contains 409,996 messages and 743,445 extracted URLs.

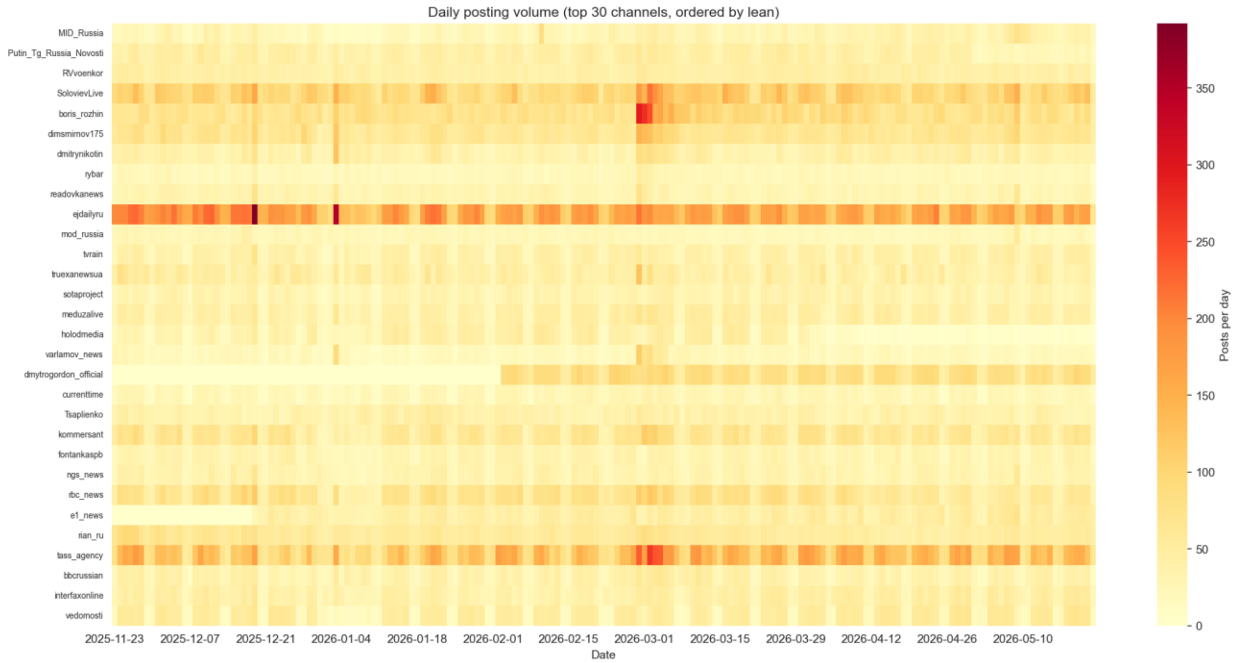

Figure 1. *Daily posting volume for the 30 most active channels over the collection window, ordered by lean. Wire services (ejdailyru, TASS) post continuously; dmytrogordon_official enters the dataset in mid-February 2026; a synchronised spike across pro-Kremlin channels is visible around early March 2026.*

2.3 Preprocessing

Text was normalized (emoji and control characters stripped, URLs separated out and kept for the co-sharing network). We removed 20,295 exact intra-channel duplicates (re-pins, repeated announcements). Cross-channel near-duplicates were detected with MinHash + LSH (Jaccard threshold 0.85), yielding 8,600 near-duplicate pairs - a direct coordination signal that later feeds the coordination score. The top copying pair (Sibir.Realii / Sever.Realii, 553 near-duplicates) are regional bureaus of the same broadcaster sharing an editorial pipeline. Channels with fewer than 100 messages were dropped (10 channels), leaving 144 active channels. URLs were canonicalized by stripping UTM / click-ID tracking parameters and fragments so that the same link counts once.

Table 1. Corpus composition by political lean.

Lean group	Messages	Share
Pro-Kremlin	214,993	52.4%
Neutral / wire	100,011	24.4%
Anti-Kremlin	88,159	21.5%
Mixed	6,833	1.7%
Total	409,996	100%

The corpus is dominated by pro-Kremlin channels. This is not a sampling artefact: state-aligned channels are both more numerous and more prolific. We control this asymmetry by reporting within-group proportions rather than raw counts throughout the content analysis.

3. Models and Methods

3.1 Network construction

We build three complementary graphs over the 144 active channels, each capturing a different relation:

- **Graph A - Forwarding (directed, weighted).** An edge A→B means channel B forwarded a post originating from A; weight = number of forwards. This is the most direct measure of information flow.
- **Graph B - URL co-sharing (undirected, weighted).** An edge means two channels posted the same external URL (platform domains such as t.me, YouTube and Twitter are excluded). This reveals latent coordination even when channels never forward each other.
- **Graph C - Temporal co-posting (undirected, weighted).** An edge means two channels posted the same URL within a 30-minute window - a signal of synchronized publishing.

3.2 SNA metrics and null models

On the forwarding graph we computed node-level centralities (weighted in/out-degree, PageRank, betweenness, eigenvector, clustering coefficient) and graph-level statistics (density, reciprocity, components, diameter, average path length, degree assortativity). To test whether the observed structure is non-random we generated two null models with matched size - Erdős–Rényi (uniform random) and Barabási–Albert (preferential attachment) - and compared clustering and degree distributions. We additionally fit a power law to the degree distribution.

3.3 Community detection and coordination

Communities were detected with Louvain (modularity optimization), Infomap (information-theoretic, via igraph), and k-clique percolation (overlapping). Detected partitions were validated against the manual lean labels using Normalized Mutual Information (NMI). For CIB detection we defined a composite coordination score for every channel pair as a weighted sum of four similarities - forwarding, URL overlap (Jaccard), temporal synchrony, and textual near-duplication - with weights 0.3 / 0.3 / 0.3 / 0.1 (the textual term is down-weighted as the noisiest).

3.4 Content analysis pipeline

NLP was pre-computed by a separate pipeline (nlp_pipeline.py) and stored in its own SQLite.The models used:

- **Sentiment -** blanchefort/rubert-base-cased-sentiment-rusentiment (RuBERT, three-way positive / negative / neutral).
- **Emotion -** cointegrated/rubert-tiny2-cedr-emotion-detection (joy, sadness, anger, fear, surprise, no_emotion).
- **NER -** spaCy ru_core_news_lg, extracting PER / ORG / LOC mentions.

On top of these we computed aspect-based sentiment (average sentiment of messages mentioning each major entity), built entity co-occurrence networks per faction, ran BERTopic (rubert-tiny2 embeddings + UMAP + HDBSCAN) on a 30K-message sample, and produced TF-IDF-weighted word clouds. Finally, as a qualitative “LLM-as-judge” layer, a local quantised model (qwen2.5:7b-instruct) annotated a 50-message sample for propaganda techniques and framing.

4. Results

4.1 Network structure (RQ1, RQ4)

The forwarding network is sharply polarised. Pro-Kremlin channels form a dense, mutually-reinforcing core, while anti-Kremlin channels occupy a separate region of the graph; the regional bureaus of the same opposition broadcaster (severrealii, sibrealii, idelrealii) are visibly isolated, forwarding almost exclusively among themselves.

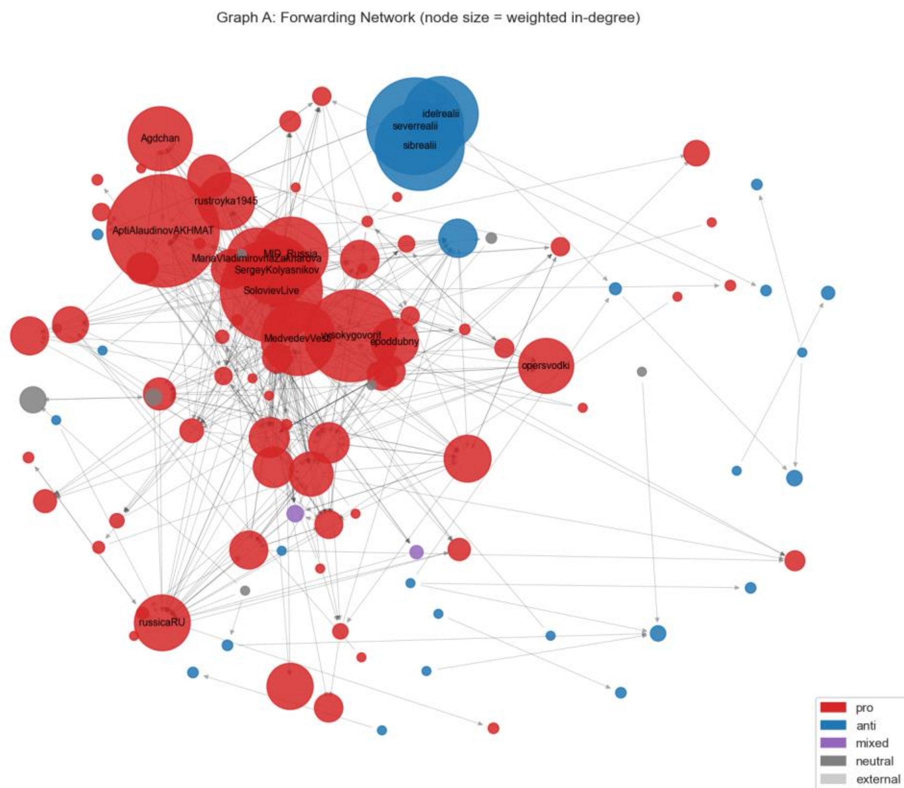

Figure 2. *Forwarding network (Graph A).*

The top channels by PageRank are all pro-Kremlin (Alaudinov, Solovyov, Obraz Budushchego, Nezygar, Medvedev), reflecting heavy mutual forwarding inside the pro core. Betweenness centrality, which highlights bridges, is led by Solovyov - a television host referenced by military bloggers, analysts and officials alike.

Graph-level statistics confirm a complex, non-random network. Density is low (0.022) but the largest weakly-connected component spans 104 of 144 channels; reciprocity is 0.25 (a quarter of forwarding ties are mutual, indicating editorial partnerships); degree assortativity is negative across all three graphs (hubs connect to low-degree periphery, a broadcast pattern).

Table 2. Real forwarding network vs. matched null models.

Metric	Real (A)	Erdős–Rényi	Barabási–Albert
Nodes / edges	144 / 463	144 / 434	144 / 423
Density	0.022	0.021	0.041
Clustering coeff.	0.251	0.046	0.132

Metric	Real (A)	Erdős–Rényi	Barabási–Albert
Max degree	28	8	38

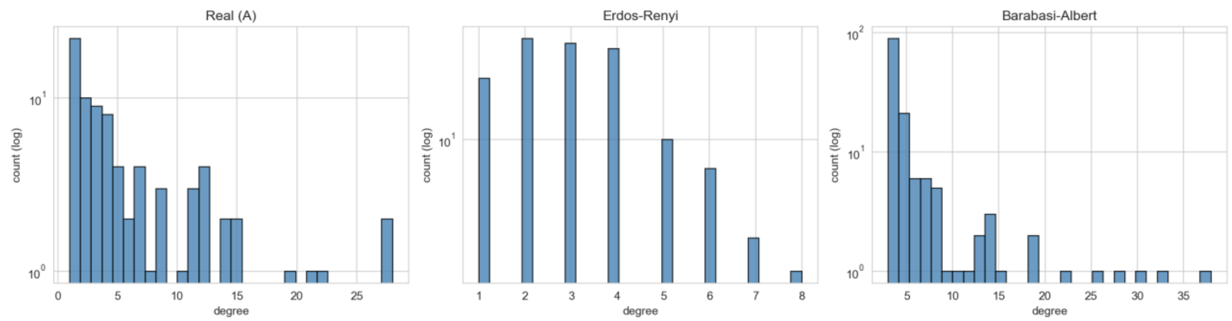

Figure 3. *Degree distributions: the real network (left) is heavy-tailed like Barabási–Albert (right) and unlike the bell-shaped Erdős–Rényi graph (centre).*

The decisive result is clustering: the real network (0.251) is roughly five times more clustered than Erdős–Rényi (0.046) and about twice Barabási–Albert (0.132). Channels form tight in-groups where everyone forwards everyone. The degree distribution is clearly heavy-tailed (fitted exponent $\alpha \approx 2.1$); the formal power-law-versus-exponential test is inconclusive at this network size (n = 144), but the structure is plainly hub-dominated and approximately scale-free.

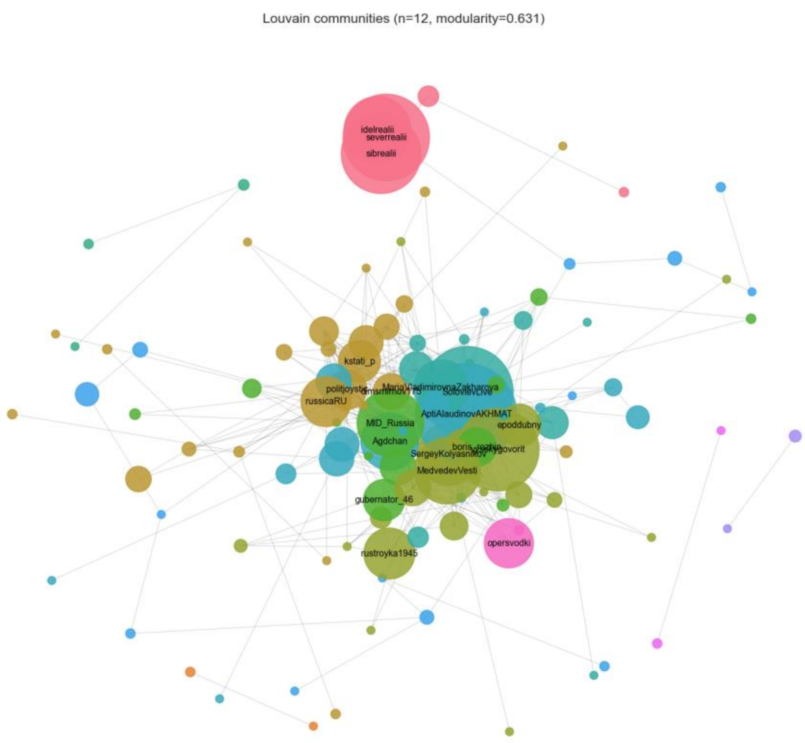

Figure 4. *Louvain community structure on the forwarding network (modularity 0.63). Colours denote detected communities; node size = degree.*

Louvain finds 43 communities at a high modularity of 0.63 (Infomap: 49 communities, in close agreement). Validated against the manual labels, NMI is 0.36 for both methods. This intermediate value is itself informative: political lean explains part of the structure, but the algorithms also recover finer sub-clusters. Pro-Kremlin channels split into military bloggers, state officials and political analysts; anti-Kremlin channels split by media affiliation. In other words, community detection captures meaningful sub-structure beyond the coarse three-way labelling.

Bridges between bubbles (RQ4) are best identified by betweenness and visualized as ego networks. Solovyov has the densest neighborhood but it is almost entirely pro - he is a hub within the pro cluster rather than a true cross-community bridge. The genuine bridge is Politjoystic, whose ego network contains pro, anti, neutral and mixed neighbours, consistent with its role as an aggregating commentary channel. Zakharova’s ego network is small and exclusively pro, matching her institutional role as ministry spokesperson.

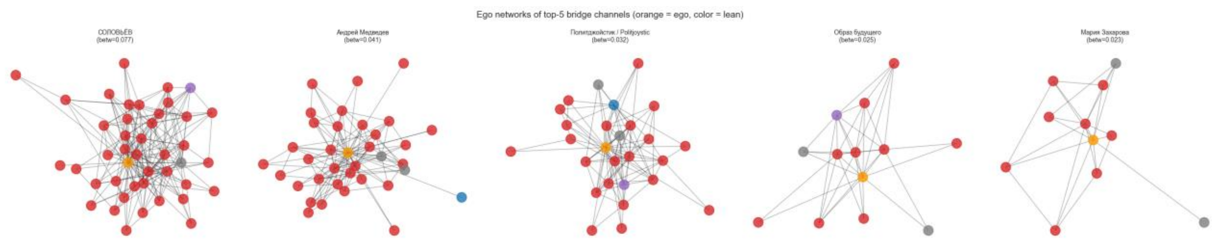

Figure 5. *Ego networks of the top-5 channels by betweenness (orange = ego). Politjoystic (centre) is the only genuinely cross-lean bridge; Solovyov (left) is a within-cluster hub.*

4.2 Coordination / CIB detection (RQ2)

The highest-scoring pairs are intuitively correct. Sever.Realii / Sibir.Realii top the ranking (0.58) as bureaus of one broadcaster; the Ministry of Foreign Affairs / Zakharova pair (0.37) reflects an institutional mirror (spokesperson echoing the ministry); the Alaudinov / Solovyov pair (0.37) links the military and media wings of the pro-Kremlin ecosystem. Almost all top pairs share a forwarding similarity of 1.0, so the URL and temporal terms are what separate genuine coordination from incidental overlap.

Figure 6. *Pairwise coordination score for the most suspicious channels, ordered by Louvain community. Bright diagonal blocks are within-community coordination; off-diagonal bright cells flag cross-community coordination worth investigating.*

The heatmap shows bright within-community blocks (expected) plus a small number of off-diagonal cells that indicate cross-community coordination. Combined with Graph C (710 temporal co-posting edges within a 30-minute window) and the 8,600 near-duplicate pairs, the evidence points to organized editorial coordination concentrated inside the pro-Kremlin core.

4.3 Content: sentiment and emotion (RQ3)

About 95% of messages are classified neutral by the sentiment model - a well-known limitation of RuBERT sentiment on formal news-register text. This does not mean the channels are unbiased; it means bias is expressed through topic selection and framing rather than overt emotional words, a point we return to with the LLM layer.

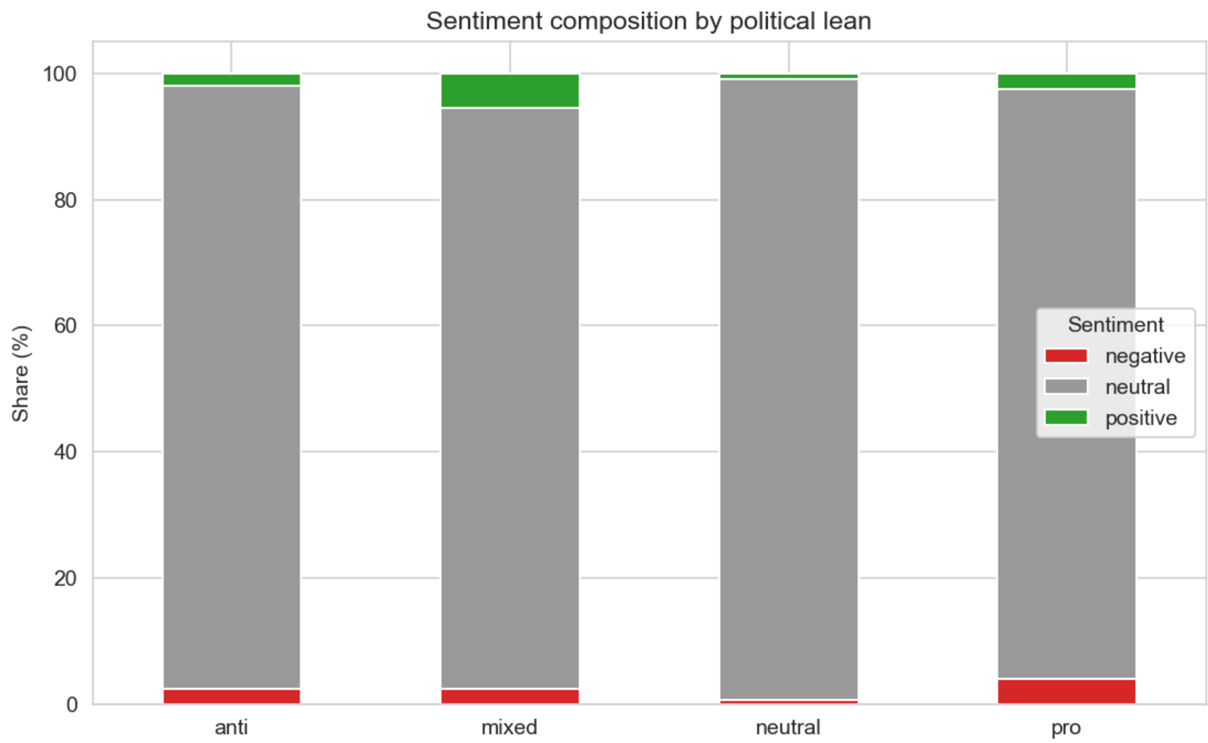

Figure 7. *Sentiment composition by lean group. Pro-Kremlin channels carry the highest negative share (4.0%); neutral wire services are almost entirely neutral.*

A counter-intuitive but robust finding emerges by lean: pro-Kremlin channels have the highest negative rate (4.0%), exceeding anti-Kremlin channels (2.4%), while neutral outlets are almost flat (0.6% negative). Pro-government channels use confrontational language about Western and Ukrainian actions; the mixed group shows the highest positive rate (5.5%), reflecting lifestyle-plus-politics channels.

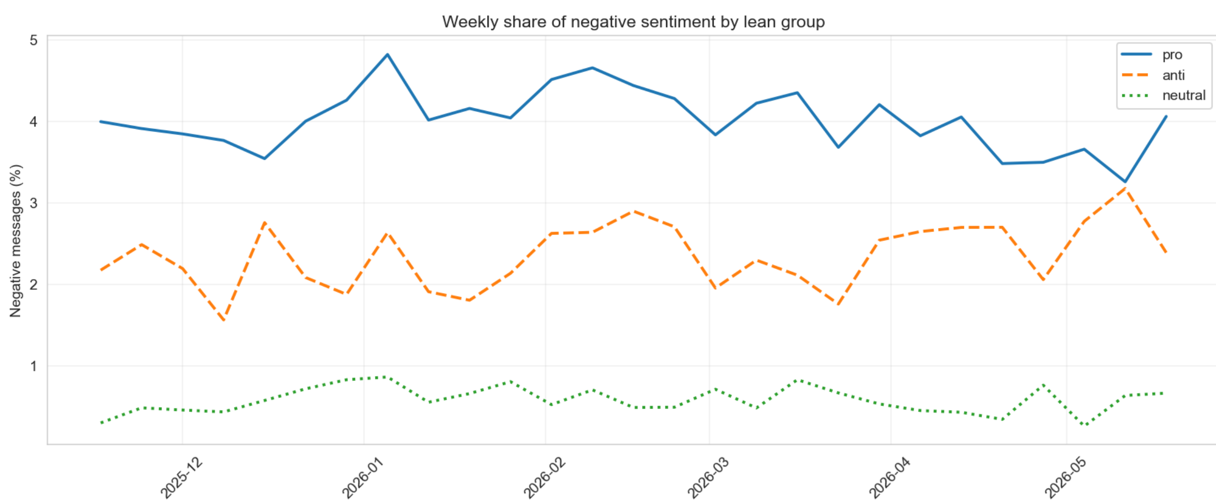

Figure 8. *Weekly share of negative sentiment by lean group. Pro-Kremlin negativity is consistently elevated and widens around military and political flashpoints.*

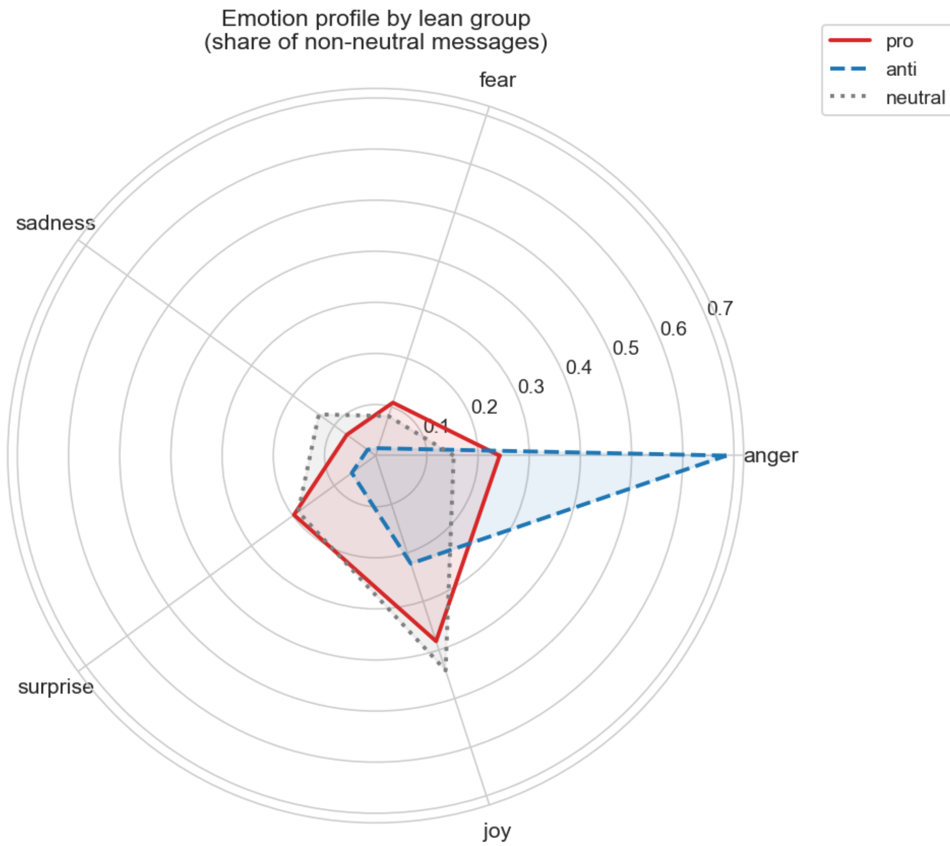

Figure 9. *Emotion profiles by lean (share of non-neutral messages). Anti-Kremlin discourse is dominated by anger; pro-Kremlin is more balanced across anger, joy and surprise.*

The emotion model leaves ~92% of messages as no_emotion; within the remaining 8% the asymmetry is striking. Anti-Kremlin channels are overwhelmingly dominated by anger (68.5%), consistent with an adversarial, critical posture. Pro-Kremlin channels are more balanced - anger (24.2%), joy (38.2%), surprise (19.7%) - alternating between aggression toward enemies and triumphalist framing of domestic achievements. Neutral outlets lean to joy and surprise, reflecting a broader editorial mix.

4.4 Content: entities and aspect-based sentiment (RQ3)

Across all three lean groups the most-mentioned person is Trump, reflecting the centrality of US–Russia relations during the collection window; Putin is second in pro channels and among the top three in anti. Among organizations, the Ukrainian Armed Forces (VSU) lead in pro channels (military reporting), whereas anti channels foreground domestic security agencies (FSB) and Western agencies (Reuters). Note that spaCy does not lemmatize Russian, so inflected forms (Trump / Trumpa) appear separately; this does not affect the co-occurrence associations.

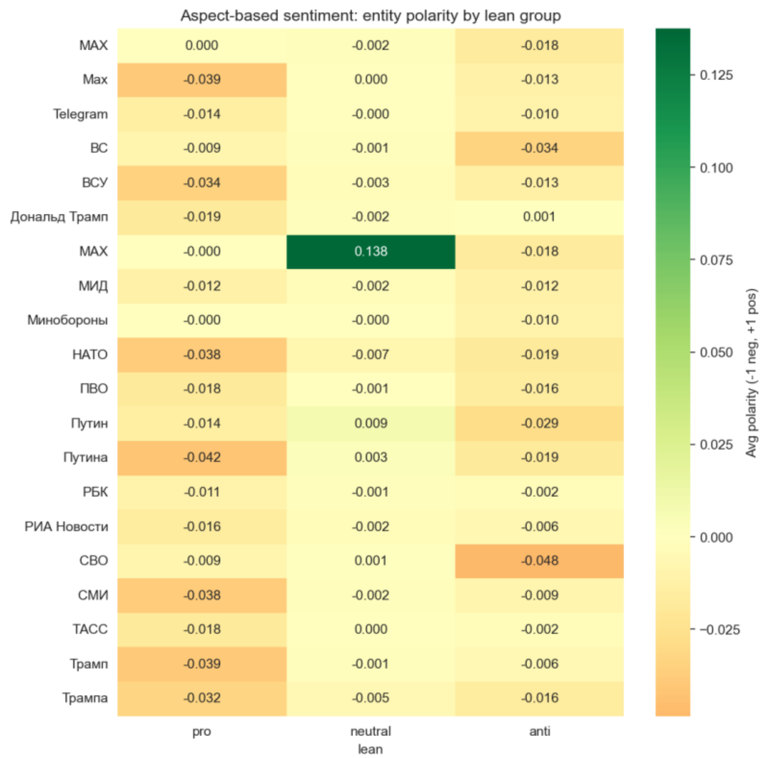

Figure 10. *Aspect-based sentiment: average polarity of messages mentioning each entity, by lean. Values are uniformly close to zero (a floor effect of the neutral-dominated model) but differ in direction between factions.*

Because overall sentiment is dominated by neutral, aspect-based polarity values are small in absolute terms — a floor effect rather than genuine neutrality. They are nonetheless directionally consistent: each side is slightly more negative toward the symbols of its out-group (anti channels toward the "special military operation" and Putin; pro channels toward VSU and NATO). This weak but coherent split is a first signature of narrative polarization, which the entity-network and topic analyses below make far clearer.

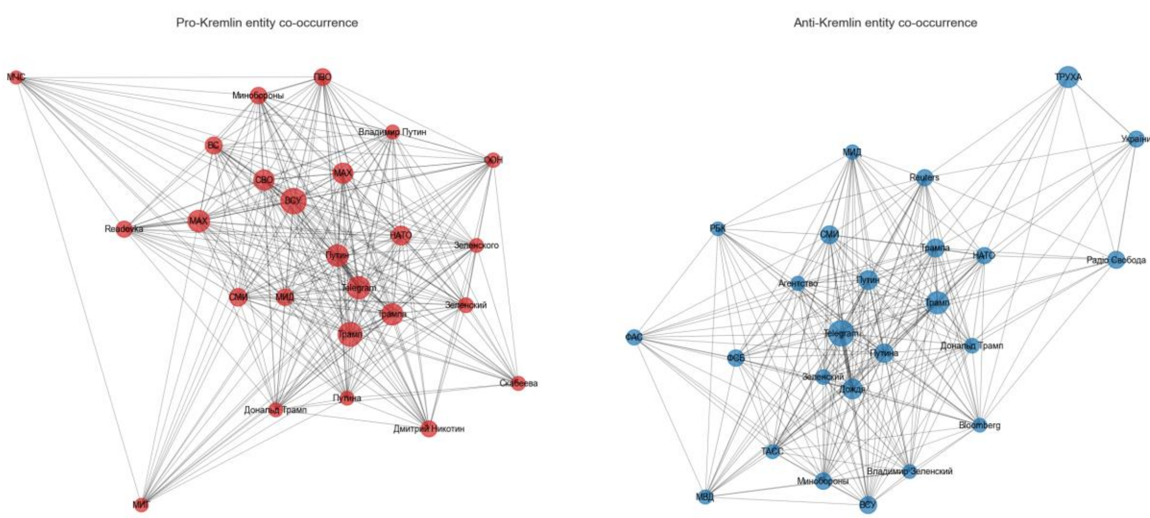

Figure 11. *Entity co-occurrence networks for pro-Kremlin (left, 280 edges) and anti-Kremlin (right, 235 edges) channels. Both center on Trump and Putin, but pro links military entities while anti foregrounds media organizations and security agencies.*

The entity co-occurrence networks make the divergence concrete. The pro-Kremlin network is denser (280 vs 235 edges), reflecting a more tightly integrated narrative; it links military entities (VSU, Akhmat, Ministry of Defence) to political figures, whereas the anti-Kremlin network gives more weight to media organizations (Reuters, Dozhd) and domestic security agencies (FSB). The same geopolitical events are narrated through different actor-networks - a visual demonstration of polarization that the flat sentiment scores conceal.

4.5 Content: topics and propaganda techniques (RQ3)

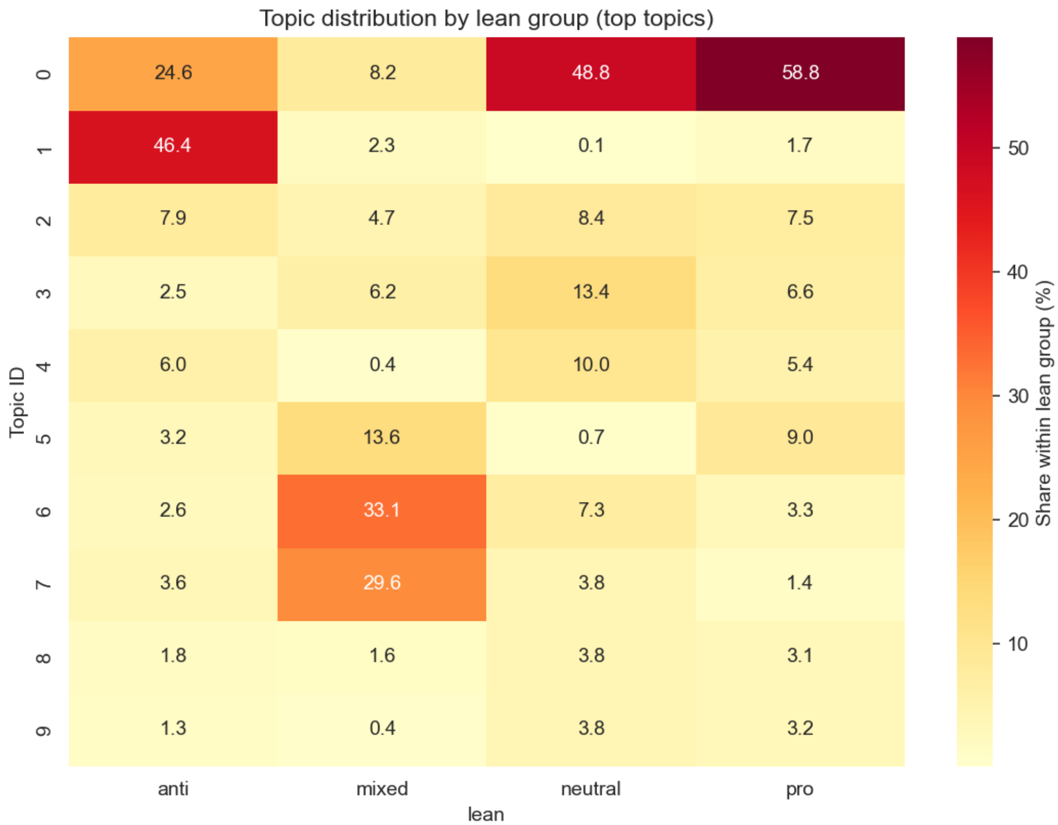

Figure 12. *Topic distribution by lean group (top BERTopic topics). Topic 0 (US–Russia–Iran geopolitics) dominates pro and neutral; Topic 1 (Ukrainian-language content) is almost exclusive to anti; Topics 6–7 (legal/financial, culture) characterize the mixed group.*

BERTopic identified 64 topics plus an outlier cluster. The largest substantive topic (Topic 0: “usa / trump / iran”) captures US–Russia–Iran geopolitics and is concentrated in pro-Kremlin (58.8%) and neutral (48.8%) channels — the shared news agenda. The most polarized topic is Topic 1 (Ukrainian-language content: “ukraina / trukha”), almost exclusive to anti-Kremlin channels (46.4%) and virtually absent from pro (1.7%), reflecting Ukrainian-language opposition outlets. The mixed group has its own signature in Topic 6 (legal / financial: rubles, court, millions — 33.1%) and Topic 7 (films / books — 29.6%), consistent with the lifestyle-plus-politics profile of controlled-opposition channels. Other distinctive topics include oil/energy markets and strikes/casualties. The topic structure thus aligns with political lean beyond the shared general-news discourse.

Figure 13. *TF-IDF-weighted word clouds by lean. Pro channels foreground military and patriotic terms; anti channels are marked by Ukrainian-language vocabulary; neutral outlets show factual reporting terms.*

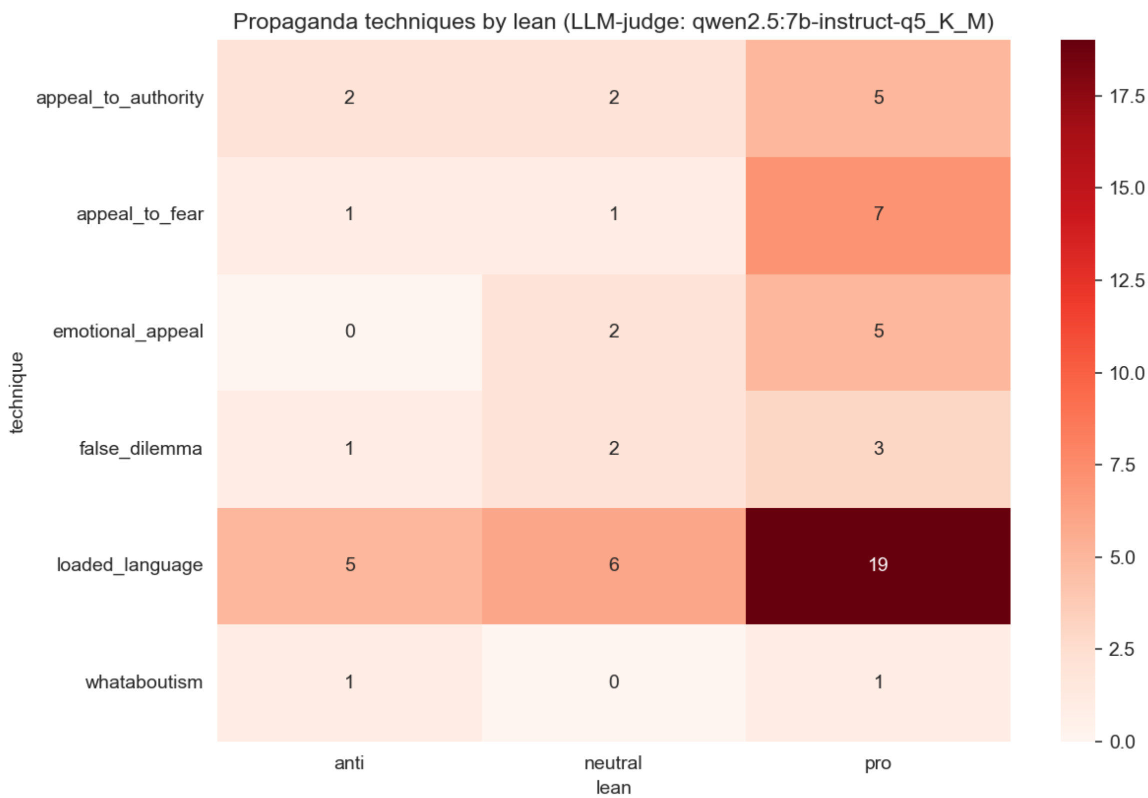

Figure 14. *Propaganda techniques detected by a local LLM judge on a 50-message sample, by lean. Loaded language and appeal to fear are concentrated in pro-Kremlin messages.*

The LLM-as-judge layer, though small (n = 50), is revealing. Loaded language is the most frequent technique and appears in pro-Kremlin channels nearly four times as often as in anti (19 vs 5); appeal to fear is almost exclusively a pro-government tool (7 vs 1). Crucially, these are the same messages the RuBERT sentiment model labelled neutral. The gap directly supports a “pseudo-neutral register” reading: state-aligned channels embed manipulation inside a formally objective tone that automated sentiment classifiers miss. The finding is a hypothesis-generator for a larger annotated sample, not a statistical claim on its own.

Figure 15. *Forwarding volume between lean groups. The dominant flow is pro→pro; cross-lean (pro↔anti) forwarding is minimal.*

Finally, aggregating forwarding by lean makes the echo-chamber structure unmistakable: the dominant flow is pro→pro, anti→anti is second and much smaller, and direct pro↔anti forwarding is negligible. Information circulates within political camps rather than across them - the macro-level confirmation of the community-detection results.

5. Discussion and Conclusions

The four research questions are answered consistently across independent methods. (RQ1) The network decomposes into a dense pro-Kremlin core and a separate anti-Kremlin region, with finer sub-communities inside each; community detection agrees with manual labels at NMI ≈ 0.36 while recovering additional structure. (RQ2) Coordination is measurable and concentrated in the pro core: high composite coordination scores, 8,600 cross-channel near-duplicates, 710 temporal co-posting ties, and institutional mirror pairs (ministry / spokesperson). (RQ3) The factions differ in emotion (anti = anger; pro = mixed aggression and triumphalism), in actor-networks (pro = military; anti = media and security agencies), and in topics (shared geopolitics vs Ukrainian-language opposition content vs mixed lifestyle/legal). (RQ4) Bridges are rare; Politjoystic is the clearest cross-lean connector, while apparent hubs such as Solovyov are bridges only within their own camp.

A recurring theme is that bias in this ecosystem is encoded structurally and lexically rather than in overt sentiment. Off-the-shelf sentiment models read the state-aligned register as neutral; only the topic, entity-network and LLM analyses expose the framing. This methodological observation - that pseudo-neutral propaganda evades sentiment classifiers - is one of the project's more transferable findings.

5.1 Limitations

- The t.me/s preview pages omit reactions and some media and truncate the history of channels with short previews (19 of 167 only partially covered).
- RuBERT sentiment and the CEDR emotion model assign most messages to the neutral / no-emotion class on news-register text, compressing dynamic range.
- spaCy NER does not lemmatize Russian inflections, so the same entity appears under several surface forms; production use would require entity linking.
- The LLM-as-judge sample (n = 50) and the overlapping-community result are illustrative rather than statistically conclusive.

Future work would add entity linking to Wikidata, scale the LLM annotation to a few thousand messages with inter-annotator agreement, and extend Graph C with content-based (not only URL-based) temporal co-posting.

6. Reproducibility and Deliverables

All code and data are organized in a single repository with a README. The pipeline is fully reproducible from the seed list:

- **wsa_scraper.py** — collects the corpus into wsa_data.db.
- **nlp_pipeline.py** — runs sentiment, emotion and NER into wsa_nlp.db (resume-safe).
- **WSA_analysis_v3.ipynb** — the main notebook: preprocessing, the three graphs, SNA metrics, community detection, coordination scoring, and the full content analysis, exporting every figure to outputs/ and the graphs to GraphML for Gephi.
- **network_builder.py** — stand-alone helper that reproduces the graphs outside the notebook.

Deliverables: this report, the collected dataset (corpus and constructed graphs), the source code with README, and the presentation.